

Problem 1 (20 Points): Let $X = (X_1, X_2)$ where

$$X_i \stackrel{iid}{\sim} \text{Exp}(\theta).$$

- i) (5 Points) Find the UMVUE for $1/\theta$.
- ii) (5 Points) Show that the geometric mean $\sqrt{X_1 X_2}$ has a smaller MSE than the UMVUE for all $\theta > 0$.
- iii) (10 Points) Find another estimator that has an even smaller MSE for all $\theta > 0$.

Problem 2 (30 Points): Let $X = (X_1, \dots, X_n)$ where

$$X_i \stackrel{iid}{\sim} f_\theta(x_i) = \exp(-(x_i - \theta))1_{\{x_i \geq \theta\}}.$$

- i) (10 Points) Find the MLE of θ and calculate its bias and MSE.
- ii) (20 Points) Find the UMREE of θ under the squared-error loss and calculate its bias and MSE. How do you compare them with those of the MLE?